

TECHNICAL DOCUMENTATION

# Yomi.

Manga & manhwa reader for Android  
Developer and user reference

Version 1.0.0 · Flutter 3.x · Dart 3.3

Platform

Android API 21+

UI

Cupertino

State

Riverpod v2

DB

Isar v3

# 1 Overview

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## 1.1 What is Yomi?

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Yomi (読 — Japanese for "reading") is a feature-complete manga and manhwa reader for Android. It provides a polished, distraction-free reading experience built entirely with Flutter's Cupertino widget set, giving it the feel of a native iOS application while running on Android hardware.

The application manages a personal library of saved titles, downloads chapters for offline reading, connects to pluggable content sources, and backs up your entire reading history to a local JSON file or Google Drive.

## 1.2 Core Design Principles

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- **Immersive, minimal UI** — Dark-first design with frosted glass overlays and tap-to-reveal chrome.
- **Reactive, no loading flickers** — Isar reactive streams + shimmer placeholders + instant Riverpod state propagation.
- **Pluggable sources** — MangaSource interface with zero build-time coupling.
- **Adaptive light & dark** — AppColorsX context extension resolves every colour at runtime.

## 2 Features

### 2.1 Library

The Library tab is the home screen. It shows every manga in a two-column grid.

- **Real-time search** — frosted search bar filters titles as you type.
- **Filter sheet** — narrow by read status (All / Unread / Read / Downloaded) and genre.
- **Categories** — user-defined shelves. Each manga can belong to multiple categories.
- **Unread badge** — displayed on the card corner when there are unread chapters.
- **Shimmer skeleton** — animated placeholder grid during database load.

### 2.2 Reader

Full-screen page view with configurable direction, scale, and background.

#### READING MODES

| Mode         | Scroll direction     | Use case                     |
|--------------|----------------------|------------------------------|
| Left → Right | Horizontal, forward  | Western comics, most manhwa  |
| Right → Left | Horizontal, reversed | Japanese manga (traditional) |
| Vertical     | Vertical             | Webtoons, long-strip manhwa  |

#### CHROME & NAVIGATION

- Tap anywhere to toggle the navigation chrome (toolbar + bottom scrubber).
- Drag or tap the scrubber track to jump instantly to any page.
- A 2 pt accent-colour progress line is always visible at the top of the screen.
- A page-count pill fades in after each page turn and disappears after 2 s.

### 2.3 Downloads

- **Queue** — chapters process one at a time with a live progress bar.
- **Pause & Resume** — any chapter can be paused; resumes from where it left off.
- **Cancel** — removes the entry; partial files are discarded.
- **Retry** — failed chapters show a retry button.

#### PER-CHAPTER STATUS INDICATORS

| Status                | Meaning                        |
|-----------------------|--------------------------------|
| Queued (clock)        | Waiting in the download queue  |
| Downloading (spinner) | Actively being downloaded      |
| Paused                | Download was manually paused   |
| Failed (X)            | Download failed — tap to retry |

Downloaded (✓)

Available offline (green tick)

## 2.4 Browse & Sources

The Browse tab lists all registered content sources.

| Source     | Language | API                                                                     |
|------------|----------|-------------------------------------------------------------------------|
| AsuraScans | English  | JSON REST — <a href="https://api.asurascans.com">api.asurascans.com</a> |
| AllManga   | English  | GraphQL — <a href="https://api.allanime.day">api.allanime.day</a>       |

## 2.5 Backup & Restore

- **Export** — creates a timestamped JSON file in app documents.
- **Restore** — merges backup into current library, preserving read/download status.
- Backup list screen with one-tap restore and confirmation dialog.

■■ Backup files are plain JSON and can be viewed or edited with any text editor.

## 2.6 Settings

| Section       | Option            | Values                            |
|---------------|-------------------|-----------------------------------|
| Appearance    | Theme             | Dark / Light                      |
| Library       | Categories        | Navigate to category management   |
| Reader        | Direction         | L→R / R→L / Vertical              |
| Reader        | Page Scale        | Fit Width / Fit Height / Original |
| Reader        | Background        | Black / White / Sepia             |
| Downloads     | Storage Location  | Local / Google Drive              |
| Extensions    | Check for Updates | Fetches latest GitHub release     |
| Backup & Sync | Export / Restore  | JSON to app documents             |

## 3 Architecture

### 3.1 Tech Stack

| Layer            | Technology                  | Version    |
|------------------|-----------------------------|------------|
| UI framework     | Flutter — Cupertino widgets | ≥ 3.3      |
| State management | Riverpod                    | ^2.5       |
| Local database   | Isar                        | ^3.1       |
| Networking       | Dio                         | ^5.4       |
| Image loading    | extended_image              | ^8.3       |
| HTML parsing     | html package                | ^0.15      |
| Backup / archive | archive                     | ^3.6       |
| Google Drive     | googleapis + google_sign_in | ^12 / ^6.2 |
| Platform bridge  | MethodChannel yomi/platform | —          |

### 3.2 Project Structure

```
lib/  
  core/  
    database/      # Isar schemas: MangaEntry, ChapterEntry, DownloadEntry  
    extensions/sources/ # MangaSource plugins (AsuraScans, AllManga, ...)  
    providers/     # Riverpod providers  
    services/     # BackupService, UpdateService  
    theme/        # AppColors, AppColorsX extension  
  features/  
    library/      # LibraryScreen, filters, category chips  
    browse/       # BrowseScreen, SourceMangaScreen  
    title_detail/ # TitleDetailScreen, ChapterListTile  
    reader/       # ReaderScreen, ReaderChrome, scrubber  
    downloads/    # DownloadsScreen  
    settings/     # SettingsScreen, BackupRestoreScreen
```

### 3.3 State Management

| Provider type         | Used for                                                                      |
|-----------------------|-------------------------------------------------------------------------------|
| StateProvider         | Simple values: brightness, reading direction, scale, background, search query |
| StreamProvider.family | Isar reactive streams — library list, chapter status, live manga object       |
| FutureProvider.family | Async loads — manga detail, chapter list, page URLs                           |

|                       |                                                                  |
|-----------------------|------------------------------------------------------------------|
| NotifierProvider      | Complex mutation — library CRUD, category management             |
| AsyncNotifierProvider | Download manager — queue, pause/resume/cancel, progress tracking |

■■ ReaderState overrides == and hashCode so that Riverpod's equality check prevents redundant rebuilds when only unrelated fields change.

## 3.4 Database Schemas (Isar v3)

| Schema        | Key fields                                                                                                 |
|---------------|------------------------------------------------------------------------------------------------------------|
| MangaEntry    | id, sourceId, sourceMangaId, title, coverUrl, author, genres, status, inLibrary, chapterCount, unreadCount |
| ChapterEntry  | id, mangaId, sourceChapterId, title, number, volume, language, uploadDate, isRead, isDownloaded            |
| DownloadEntry | id, mangaId, chapterId, mangaTitle, chapterNumber, status, progress, downloadedPages, totalPages           |

## 3.5 Theming

The AppColorsX extension on BuildContext resolves every colour at runtime:

```
// Use in any build method:
final bg      = context.backgroundColor;      // dark: #0A0A0F   light: #F2F2F7
final surface = context.surfaceColor;        // dark: #141418   light: #FFFFFF
final elevated = context.surfaceElevatedColor; // dark: #1C1C24   light: #ECECF0
final border  = context.borderColor;        // dark: 8% white  light: 8% black
final textSec = context.textSecondaryColor;  // 60% opacity
```

## 4 Source Plugin System

### 4.1 MangaSource Interface

```
abstract class MangaSource {  
  String get id;          // unique slug, e.g. "asurascans_en"  
  String get name;        // display name  
  String get baseUrl;     // root URL for web links  
  String get language;    // ISO 639-1 code  
  Map<String, String> get imageHeaders;  
  List<SourceFilter> getFilters();  
  
  Future<List<MangaSummary>> fetchPopular({int page = 1});  
  Future<List<MangaSummary>> fetchLatestUpdates({int page = 1});  
  Future<List<MangaSummary>> search(String query, ...);  
  Future<MangaDetail> fetchMangaDetail(String mangaId);  
  Future<List<ChapterInfo>> fetchChapterList(String mangaId);  
  Future<List<String>> fetchPageUrls(String chapterId);  
}
```

### 4.2 Built-in Sources

#### AsuraScans (asurascans\_en)

- API base: `https://api.asurascans.com/api`
- Chapter ID format: `seriesSlug::chapterUUID::chapterNumber`
- `fetchPageUrls`: 3-strategy fallback (UUID path → chapter-number query → direct UUID endpoint)
- Sends Origin and Referer headers with every request.

#### AllManga (all\_manga\_en)

- API base: `https://api.allanime.day/api` (GraphQL)
- Chapter ID format: `mangaId::chapterNumber`
- Tries POST first (standard GraphQL), falls back to GET on `DioException`.

### 4.3 Adding a New Source

1. Create `lib/core/extensions/sources/my_source.dart` and implement `MangaSource`.
2. Register your instance in `lib/core/providers/source_registry_provider.dart`.

■■ Image headers must include Referer and Origin — many CDNs reject requests without them.

# 5 Getting Started

## 5.1 Prerequisites

| Requirement | Version                        |
|-------------|--------------------------------|
| Flutter SDK | ≥ 3.3.0                        |
| Dart SDK    | ≥ 3.3.0                        |
| Android SDK | API 21+ (Android 5.0 Lollipop) |
| Java / JDK  | 17 (required by Gradle 8)      |

## 5.2 Build & Run

### 1. Clone and fetch dependencies

```
git clone https://github.com/tsnyders/comic-center.git
cd comic-center
flutter pub get
```

### 2. Generate code

Isar schemas and Riverpod annotations require code generation:

```
dart run build_runner build --delete-conflicting-outputs
```

■ Never manually edit \*.g.dart or \*.freezed.dart files — they are regenerated on every build\_runner run.

### 3. Run & Analyse

```
# Debug on a connected device or emulator
flutter run

# Release build (optimised APK)
flutter build apk --release

# Static analysis (expect info-level warnings only)
flutter analyze
```

■ Critical: Do NOT change the applicationId (com.comiccenter.comic\_center). Changing it forces users to uninstall and reinstall, losing all local data.



## 6 Settings Reference

All settings are persisted via Riverpod StateProviders backed by SharedPreferences. Changes take effect immediately.

### Reader

| Setting           | Provider                 | Options                         | Default  |
|-------------------|--------------------------|---------------------------------|----------|
| Theme             | brightnessProvider       | dark / light                    | dark     |
| Reading Direction | readingDirectionProvider | ltr / rtl / vertical            | ltr      |
| Page Scale        | pageScaleModeProvider    | fitWidth / fitHeight / original | fitWidth |
| Background        | readerBackgroundProvider | black / white / sepia           | black    |
| Download Location | downloadLocationProvider | local / googleDrive             | local    |

### Update Check

Calls: <https://api.github.com/repos/tsnyders/comic-center/releases/latest>

- **HTTP 200** — release exists. Displays version tag and changelog body. Download button if APK asset is attached.
- **HTTP 404** — no releases yet. Shows "No Updates Available".
- **Network error** — throws UpdateCheckException. Shows "Error" with specific message.

# 7 Backup & Data

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## Backup File Format

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Backups are stored as UTF-8 JSON at:

```
{appDocumentsDirectory}/backups/yomi_backup_YYYYMMDD_HHmss.json
```

Top-level structure:

```
{
  "version": 1,
  "exportedAt": "2026-06-07T10:30:00Z",
  "categories": ["Favourites", "Reading"],
  "manga": [
    {
      "title": "Solo Leveling",
      "sourceId": "asurascans_en",
      "sourceMangaId": "solo-leveling-abc123",
      "categories": ["Favourites"],
      "chapters": [
        { "number": 1, "isRead": true, "isDownloaded": false }
      ]
    }
  ]
}
```

## Restore Behaviour

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The restore operation performs an upsert merge:

- Find the existing record by sourceId + sourceMangaId.
- If found, update metadata — **preserve** isRead and isDownloaded on chapters.
- If not found, insert as a new entry.
- Categories in the backup are merged with existing ones (no duplicates).

■■ A restore never deletes existing data. To start fresh, clear app data from Android Settings first.

## Pull-to-Refresh

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Pulling down on the chapter list triggers refreshMangaChapters:

- New chapters are inserted.
- Existing chapters have title, number, and uploadDate updated.
- isRead and isDownloaded are **never** modified.
- chapterCount and unreadCount on MangaEntry are recalculated.